

Removing the Poincaré Hypothesis for Sobolev Data with Bounded Lower Jet

Candidate partial result for arXiv:1801.02824

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Status: candidate partial result, pending human review. For every bounded simply connected planar domain and every finite p , the Poincaré assumption can be removed for the subspace $W^{k,p} \cap W^{k-1,\infty}$. The unrestricted density question remains open.

1 Source question and partial answer

Nandi, Rajala, and Schultz prove that $W^{k,\infty}(\Omega) \cap C^\infty(\Omega)$ is dense in the homogeneous space $L^{k,p}(\Omega)$ whenever $\Omega \subset \mathbb{R}^2$ is bounded and simply connected [1]. They obtain density in the full $W^{k,p}$ norm when Ω is additionally a p -Poincaré domain, and ask whether this extra assumption can be dropped (PDF page 2, after Corollary 1.2).

sequence. we achieve this via an elementary use of simply connectedness in the plane. in both of these approaches the values of the function in a suitable compact set are used to define a smooth function in the entire domain which approximates the original function in Sobolev norm. For this we employ similar tools as used by Jones in [8].

In p -Poincaré domains, that is domains Ω where a p -Poincaré inequality

$$\int_{\Omega} |u - u_D|^p dx \leq C \int_{\Omega} |\nabla u|^p dx$$

holds, we can bound the integrals of the lower order derivatives by the integrals of the higher order ones and thus we obtain the following corollary to our Theorem 1.1

Corollary 1.2. *Let $k \in \mathbb{N}$, $p \in [1, \infty)$ and $\Omega \subset \mathbb{R}^2$ be a bounded simply connected p -Poincaré domain. Then $W^{k,\infty}(\Omega) \cap C^\infty(\Omega)$ is dense in the space $W^{k,p}(\Omega)$.*

For instance Hölder-domains are p -Poincaré domains for $p \geq 2$, see Smith-Stegenga [21]. It still remains an open question whether Corollary 1.2 holds if one drops the assumption of being a p -Poincaré domain.

Next we come to the question of density of $C^\infty(\mathbb{R}^2)$ functions in $L^{k,p}(\Omega)$ in our setting of bounded simply connected domains. We have the following corollary which is analogous to

Figure 1: The source question in arXiv:1801.02824, PDF page 2.

Here and below $W^{0,\infty} = L^\infty$. We prove the following proper subcase of the question.

Theorem 1 (bounded-lower-jet density). *Let $k \in \mathbb{N}$, $1 \leq p < \infty$, and let $\Omega \subset \mathbb{R}^2$ be a bounded simply connected domain. Then*

$$W^{k,\infty}(\Omega) \cap C^\infty(\Omega) \quad \text{is dense in} \quad W^{k,p}(\Omega) \cap W^{k-1,\infty}(\Omega)$$

with respect to the $W^{k,p}(\Omega)$ norm.

For $k = 1$ this recovers only the bounded-data reduction inside the known full first-order theorem. The new content is for $k \geq 2$: no Poincaré, extension-domain, or boundary regularity assumption is imposed.

2 Proof intuition

The source approximant keeps the function on a compact Whitney core and puts one moment-normalized polynomial on each complementary boundary tail. At order k , the polynomial itself disappears and only differences of nearby polynomials remain; this is why the published proof is homogeneous. At a lower order, the polynomial survives on the entire tail and can be large far from its anchoring square.

Boundedness of the complete lower jet neutralizes precisely this failure. The moment conditions then bound every coefficient of every anchoring polynomial, uniformly in the Whitney scale. Since the domain is bounded, all these polynomials and their lower derivatives are uniformly bounded on the whole domain. Their remaining L^p contribution is supported in a boundary tail whose measure tends to zero. A variable-radius interior smoothing lemma lets us make the datum smooth without losing the bounded lower jet.

3 Two lemmas

For a square $Q \Subset \Omega$, let $P_Q = P_k(u, Q)$ be the polynomial of degree at most $k - 1$ determined by

$$\int_Q D^\alpha (u - P_Q) = 0, \quad |\alpha| \leq k - 1. \quad (1)$$

This is the normalization used in [1].

Lemma 2 (uniform polynomial extrapolation). *Suppose $u \in W^{k-1,\infty}(\Omega)$ and P_Q is defined by (1). If $d = \text{diam } \Omega$, then, for $0 \leq m < k$,*

$$\|D^m P_Q\|_{L^\infty(\Omega)} \leq C(k, d) \max_{|\gamma| \leq k-1} \|D^\gamma u\|_{L^\infty(\Omega)}. \quad (2)$$

The constant is independent of Q .

Proof. Let c_Q and $\ell(Q)$ be the center and side length of Q . Write P_Q in Taylor form about c_Q . For $|\gamma| = k - 1$, (1) says that the constant $D^\gamma P_Q$ is the average of $D^\gamma u$ on Q , so its coefficient is bounded by the right side of (2). Descend on $|\gamma|$. The average $|Q|^{-1} \int_Q D^\gamma P_Q$ equals $|Q|^{-1} \int_Q D^\gamma u$, while its difference from $D^\gamma P_Q(c_Q)$ is a fixed linear combination of already controlled higher coefficients multiplied by powers of $\ell(Q)$. Since $\ell(Q) \leq d$, every Taylor coefficient is bounded by $C(k, d)$ times the right side of (2). Finally $|x - c_Q| \leq d$ for $x \in \Omega$, and evaluation of the Taylor polynomial gives (2). $\square$

Lemma 3 (bounded-jet interior smoothing). *For every $u \in W^{k,p}(\Omega) \cap W^{k-1,\infty}(\Omega)$ there are $v_j \in C^\infty(\Omega) \cap W^{k,p}(\Omega) \cap W^{k-1,\infty}(\Omega)$ such that $v_j \rightarrow u$ in $W^{k,p}(\Omega)$.*

Proof. Choose a positive $r \in C^\infty(\Omega)$ such that $r(x) < \text{dist}(x, \partial\Omega)/4$ and every derivative of r is bounded. Such an r is obtained by composing a regularized distance with a positive smooth function flat at zero and then rescaling. For a standard mollifier ρ supported in the unit disk, set

$$v_\eta(x) = \int_{B(0,1)} \rho(y) u(x - \eta r(x)y) dy. \quad (3)$$

For small η , all sampled points lie in Ω , and the smooth kernel form of (3) shows that $v_\eta \in C^\infty(\Omega)$.

For fixed y , $F_{\eta,y}(x) = x - \eta r(x)y$ is uniformly bi-Lipschitz, $F_{\eta,y} \rightarrow \text{id}$ uniformly, and its Jacobian tends uniformly to one. Hence $f \circ F_{\eta,y} \rightarrow f$ in $L^p(\Omega)$ for every $f \in L^p$, uniformly for $|y| \leq 1$ after the usual density argument. Differentiating (3) weakly through order k , the principal term at order m is the average of $(D^m u) \circ F_{\eta,y}$ multiplied by tensors tending uniformly to the identity. Every other term contains a derivative of ηr and a derivative of u of order at most $m-1$. Those coefficients are $O(\eta)$; for $m \leq k$ the accompanying lower derivative is in L^p , and for $m \leq k-1$ it is also bounded. Dominated convergence therefore gives $v_\eta \rightarrow u$ in $W^{k,p}$, while the same formula gives $\|v_\eta\|_{W^{k-1,\infty}} < \infty$ for every fixed η . Take $\eta \downarrow 0$. $\square$

4 Proof of Theorem 1

Proof. By Lemma 3 and a diagonal argument, it is enough to treat

$$u \in C^\infty(\Omega) \cap W^{k,p}(\Omega) \cap W^{k-1,\infty}(\Omega). \quad (4)$$

Use exactly the Whitney cores D_n , boundary regions H_i , partition $(\psi_i)_{i=0}^{l_n}$, associated squares Q_i , and polynomials $P_i = P_{Q_i}$ from Sections 2–4 of [1], and put

$$u_n = u\psi_0 + \sum_{i=1}^{l_n} \psi_i P_i. \quad (5)$$

The source proves that $u_n \in C^\infty(\Omega) \cap W^{k,\infty}(\Omega)$, $u_n = u$ on D_{n-1} , and

$$\|D^k(u_n - u)\|_{L^p(\Omega)} \rightarrow 0. \quad (6)$$

We record what the same calculation gives below top order. Let $R_n = \Omega \setminus D_{n-1}$ and let R_n^* denote the union of the bounded-overlap boundary-square chains used in equations (4.3)–(4.8) of [1]. Then R_n, R_n^* escape every compact subset of Ω . For $0 \leq m < k$, Leibniz' rule, the cutoff bounds $|D^a \psi_i| \lesssim 2^{n|a|}$, the local polynomial Poincaré estimate, and the bounded-length chaining lemma of the source give

$$\begin{aligned} \|D^m(u_n - u)\|_{L^p(\Omega)} &\leq C\|D^m u\|_{L^p(R_n)} + C\left(\sum_i \|D^m P_i\|_{L^p(H_i)}^p\right)^{1/p} \\ &\quad + C2^{-n(k-m)}\|D^k u\|_{L^p(R_n^*)}. \end{aligned} \quad (7)$$

Indeed, the last term contains all products in which a derivative lands on a cutoff: after subtracting a neighboring polynomial, the chaining estimate contributes $2^{-n(k-|\beta|)}$, while the cutoff contributes $2^{n(m-|\beta|)}$. The only terms not covered by this cancellation are the plain $D^m P_i$ terms displayed explicitly in (7).

Set $M = \max_{|\gamma| \leq k-1} \|D^\gamma u\|_{L^\infty(\Omega)}$. The regions H_i have overlap bounded independently of n . Lemma 2 therefore yields

$$\left(\sum_i \|D^m P_i\|_{L^p(H_i)}^p\right)^{1/p} \leq C(k, \Omega) M |R_n|^{1/p}. \quad (8)$$

The exhaustion $D_n \uparrow \Omega$ implies $|R_n| \rightarrow 0$ and absolute continuity of the integral gives $\|D^m u\|_{L^p(R_n)} \rightarrow 0$. The final term in (7) is bounded by $C2^{-n(k-m)}\|D^k u\|_{L^p(\Omega)}$ and also tends to zero. Equations (6)–(8), for every $m < k$, prove $u_n \rightarrow u$ in $W^{k,p}(\Omega)$. $\square$

5 Scope and status

The theorem does not resolve the source question for arbitrary $W^{k,p}$ data. Without a bounded lower jet, (8) is unavailable: a polynomial fixed on a tiny associated square may extrapolate a large lower-order value throughout a geometrically large tail. This is exactly the obstruction that the global Poincaré inequality suppresses in the source corollary.

Eight focused upgrade attempts tested direct lower-order estimates, abstract jet density, room-and-corridor counterexamples, higher-order truncation, intrinsic compatibility, and adaptive Whitney trees. No full proof or counterexample survived. Bounded exact-phrases, citations, titles, and author searches through 13 August 2026 found no later resolution. The 2021 Gromov-hyperbolic generalization remains homogeneous and retains a Poincaré hypothesis for the full norm [2]. The candidate theorem appears new, but novelty confidence is only moderate pending expert literature review.

Human-review recommendation. Check the lower-order bookkeeping leading to (7), especially the bounded overlap of the regions used for the plain-polynomial term, and check the weak chain rule in Lemma 3. These are the two load-bearing points.

References

- [1] D. Nandi, T. Rajala, and T. Schultz, *A density result for homogeneous Sobolev spaces on planar domains*, Potential Analysis 51 (2019), 483–498; arXiv:1801.02824.
- [2] D. Nandi, *A density result for homogeneous Sobolev spaces*, Nonlinear Analysis 213 (2021), 112501; arXiv:1803.08715.